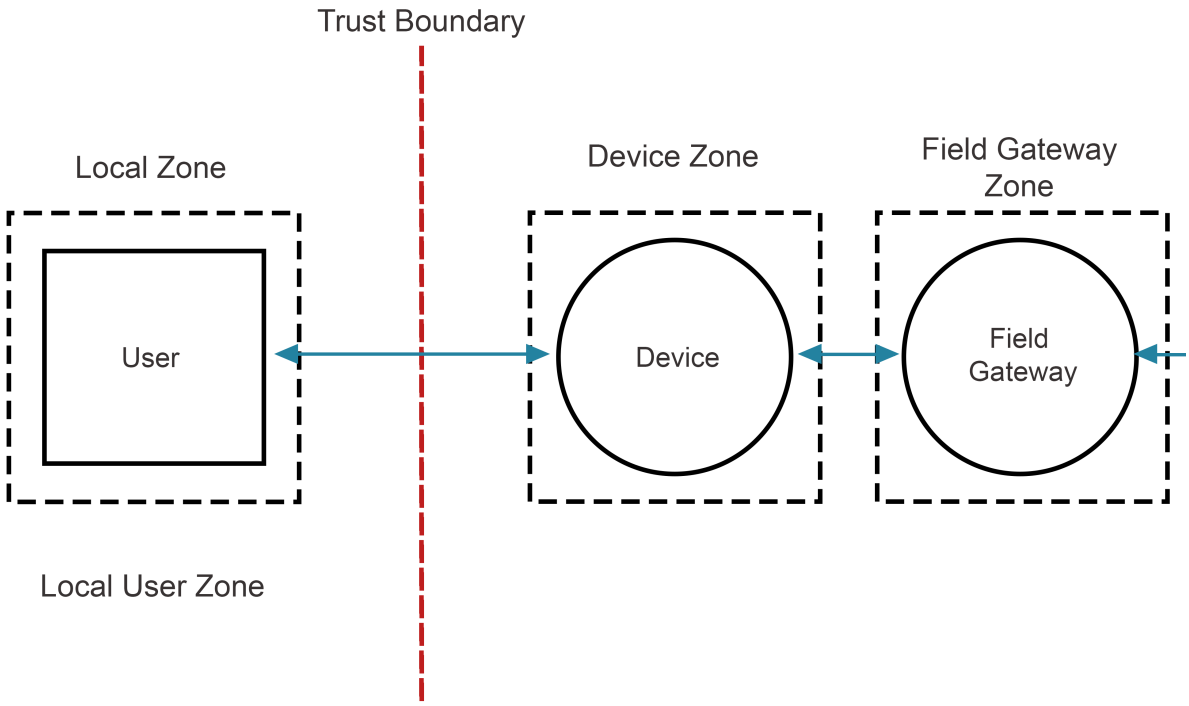
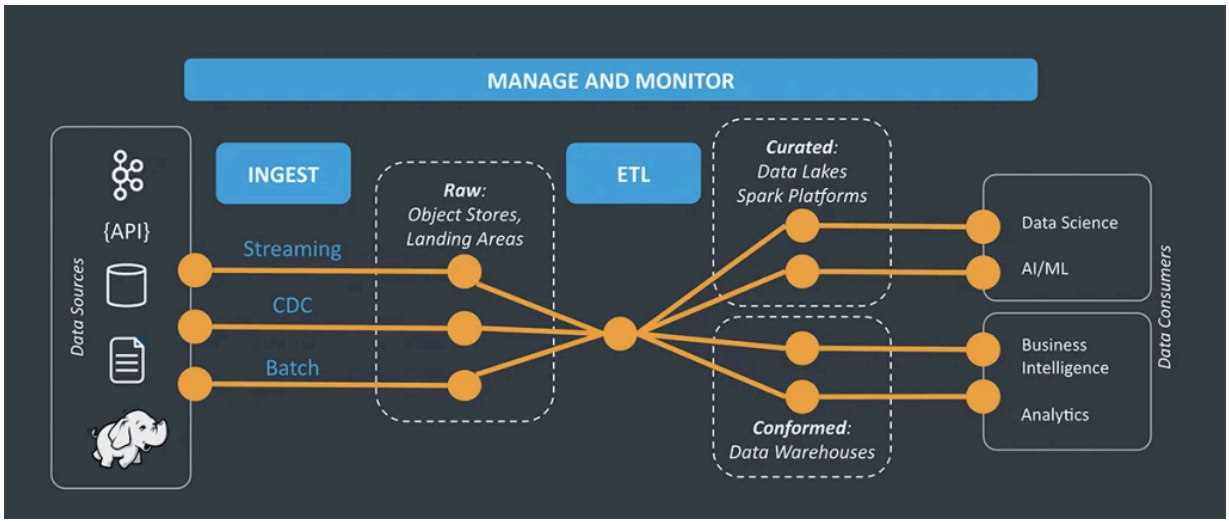
MODULE 1 — Data Ingestion: How AI Thinks About Inputs

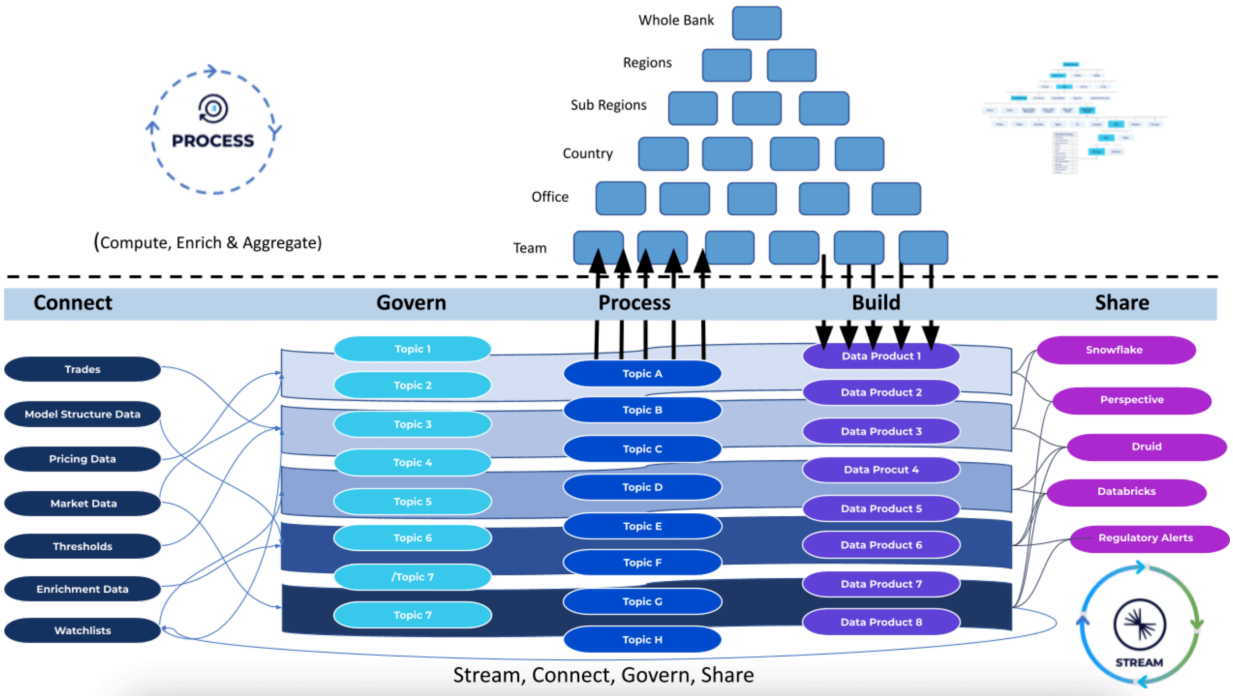
Trust Calibration, Not File Loading



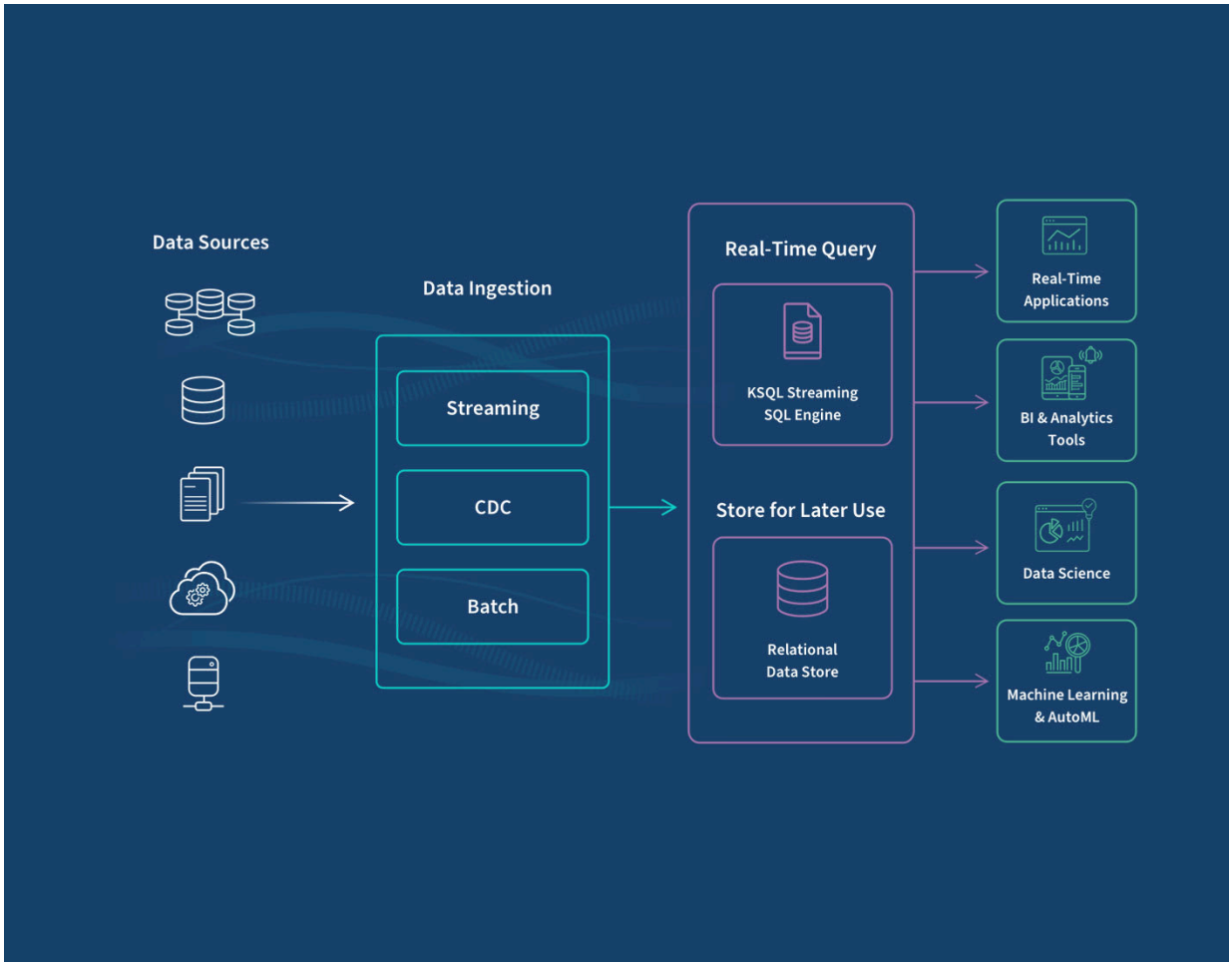
https://appcheck-ng.com/what-is-a-trust-boundary-and-how-can-i-apply-the-principle-to-improve-security/?utm_source=chatgpt.com



https://estuary.dev/blog/data-ingestion-pipeline/?utm_source=chatgpt.com



https://www.confluent.io/blog/real-time-risk-analysis/?utm_source=chatgpt.com



https://www.qlik.com/us/data-ingestion?utm_source=chatgpt.com

Why This Module Matters

Humans treat ingestion as **retrieval**. AI treats ingestion as **trust calibration**.

Once data crosses the ingestion boundary, AI implicitly assumes:

- this data is *authoritative enough* to influence a decision
- its limitations are *acceptable*
- its failure modes are *understood*

If those assumptions are wrong, **every downstream step is compromised**, no matter how good the model is.

Anchor Back to Our Decision (Always Do This)

From Module 0, we decided:

We are deciding which customer cohorts should receive churn interventions under budget and operational constraints, with uncertainty about whether recent behavioral changes reflect true

churn risk or temporary noise.
Ingestion sets the ceiling on how well this decision can be made.
AI immediately asks:
"How much should I trust this evidence?"

Core Reframe

To AI, ingestion is not about loading data — it is about establishing how much belief downstream decisions are allowed to place in that data.
Bad ingestion doesn't just cause errors. It causes **overconfidence**.

Takeaways

- AI acknowledging that no data source offers absolute truth, insights & takeaways are only as quality & trustworthy as the data that is available to act on.

AI's Ingestion Decision Questions

Before *any* cleaning or modeling, AI implicitly asks:

1. **Is this source authoritative, proxy, or heuristic?**
2. **What decisions would break if this data is wrong?**
3. **What failure modes exist at the boundary?**
4. **Is ingestion batch, streaming, or episodic — and why?**

If you cannot answer these, ingestion is incomplete.

Types of Inputs (How AI Categorizes Them)

AI mentally tags inputs like this:

- **Authoritative** Ground truth systems (billing, contracts, audited systems)
  - Example: Transaction data (e.g., Amazon purchases)
- **Proxy** Behavioral signals that *approximate* intent (usage, clicks, support tickets)
  - Example: website traffic
- **Heuristic** Derived or inferred signals (scores, flags, weak labels)
  - Example: churn risk score
  - Note: "If the data already answers part of the question you're trying to decide, it's probably heuristic."
  - Labels are not only inferred, they're **post-hoc judgements frozen in time**.

Appendix

- Under Module 1, see *Heuristic Data Source - What AI Means by This*

🔑 Most churn signals are **proxies**, not truth — and AI must be told that explicitly.

What Often Goes Wrong (Silent Failure Patterns)

- Treating proxy data as authoritative
- Ignoring label delay or label noise
  - Label delay -> churn flag hasn't been assignment because event hasn't occurred
  - Label noise -> administrator flagged an event as "churned" as opposed to event being observed naturally
  - **Outcome: both lead to AI confidently optimizing against incorrect truth signals unless uncertainty is made explicit**
- Assuming historical churn labels are “correct”
- Missing changes in upstream collection logic
- Letting ingestion quietly change without surfacing risk

None of these throw errors. All of them distort decisions.

Prompt Engineering Pattern (You’ll Reuse This)

When working with AI at ingestion time, use:
“Given this data source, list assumptions, failure modes, and what decisions break if this data is wrong.”
This forces AI to reason about **risk**, not convenience.

Takeaways

- Ingestion is all about providing AI with the correct assumptions/context about each data source identified as being useful to addressing some business problem. -> NOT telling AI what data sources to use & how to use them
  - It helps AI determine what data sources to rely on in approach (decision-making) & how by how much (trust)
- **Big picture:** ingestion != data loading, EDA, etc.
- When identifying failure, be **specific** rather than **referential** -> reinstate what was previously discussed **-> AI explicitly needs this as part of weighing decision paths**
  - **Referential:** "If this data is wrong or misleading, the most dangerous downstream decision failure would be action taken resulting in one of the previously identified, irreversible outcomes."

- **Specific:** "If this data is wrong or misleading, the most dangerous downstream decision failure would be systematically intervening on the wrong customer cohorts, incurring irreversible costs while failing to prevent churn in the truly at-risk population."

🧪 Module 1 Mini-Exercise (Let's Do It Together)

Step 1 — Pick One Input Source (from our scenario)

Choose **one**:

- customer usage logs
- support ticket volume
- billing / tenure data
- historical churn labels

(You can pick whichever feels most interesting.)

Step 2 — Classify the Source

Answer briefly:

- Is it **authoritative**, **proxy**, or **heuristic**?
- Who controls it?
- How confident *should* downstream decisions be?

Step 3 — Boundary Failure Thought Experiment

Answer this sentence:

"If this data is wrong or misleading, the most dangerous downstream decision failure would be _____."

No fixes yet. Just exposure.

Rules (Important)

❌ Do not clean, transform, or model
❌ Do not propose solutions
✅ Only reason about **trust, assumptions, and breakage**

When you're ready, reply with:

- which input you chose
- your classification
- your failure sentence

I'll review it the same way — as a **senior AI-native ingestion reviewer** — and then we'll move naturally into **Module 2: Data Cleaning (Bias Shaping, Not Fixing)**.
Take your time. This is where good systems start.

Excellent — this is where data science starts to feel **uncomfortable in a productive way**.